

Aarav Tiwari

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EDUCATION

University of North Carolina at Chapel Hill

Chapel Hill, NC

B.S. in Computer Science & Statistics and Analytics — GPA: 3.88 / 4.0

Expected May 2028

- Relevant Coursework: Algorithms & Analysis, Linear Algebra for Applications, Intro Machine Learning, Systems Programming, Data Structures, Computer Organization, Methods of Data Science

EXPERIENCE

Software Engineering Intern

Durham, NC

Fidelity Investments

June 2026 - Aug 2026

- Migrated the Tiles frontend across **three major Mantine versions (v6→v9)**, refactoring **120+** components onto **CSS-variable theming**; removing the Emotion Babel transform cut bundle size **25%** and build times **15%**.
- Architected a drag-and-drop layout builder (**react-draggable/react-resizable**), mapping live internal REST data into modular “Tiles”, with per-user layout persistence to **PostgreSQL** and **session-based RBAC** via **next-auth**, gating create/edit permissions.
- Deployed the platform on **AWS** as Fidelity’s in-house signage solution across **14 regional campus sites**, migrating off a licensed third-party vendor, allowing associates to create their own layouts with no per-site setup.

PROJECTS

MSHN | [GitHub](#) | *Python, FastAPI, Claude API, Postgres/pgvector, Next.js, Docker*

Aug 2026

- Built a **retrieval-augmented** market-anomaly explainer **pgvector** that indexes SEC filings ahead of the price event, grounding each explanation in **pre-retrieved data** rather than fetching after the price movement.
- Designed an **LLM-judged eval harness** scoring outputs on four axes; **separated factual grounding from causal reasoning**, isolating causal reasoning as the failure mode and lifting its score from **0.21 to 0.75** overall.
- Cached LLM explanations in **Postgres** to serve repeat queries at **zero token cost**, capping API spend under public traffic; deployed on **Vercel and Railway**.

Stym | [GitHub](#) | *Next.js 15, tRPC, Prisma/PostgreSQL*

May 2026

- Designed a **full-stack** pharmacokinetic engine simulating per-minute absorption/decay curves for **12+** compounds with interaction modeling to predict focus/crash windows in real time.
- Built an LLM recommendation feature (**Anthropic SDK**) behind a **type-enforced** privacy boundary, guaranteeing no identifiable data reaches the model, with hashed analytics and rate-limited endpoints.
- Built a background scoring pipeline (**BullMQ + cron**) computing a 3-pillar daily performance grade, powering notifications, digests, and 90-day trend views in a **sub-3-tap mobile UI**.

Unidad | [SolHacks 2026 Winner](#) | *Next.js, Supabase, Gemini API, IBM watsonx*

March 2026

- Engineered a **3-agent AI orchestrator** (Habla, Linda, and Dante), automating complex, multi-stage government form submission for non-English speakers.
- Built a **multi-functional browser extension** to scrape DOM field selectors, inject translated data payloads, and programmatically handle automated form submissions for **3 major job sites** (Indeed, Workday, Glassdoor).
- Architected a **stateless backend pipeline** reordering and translating raw field data into validated JSON, implementing a **DB-backed hydration fallback** to ensure 100% UI state correctness.

TECHNICAL SKILLS

Languages: Java, Python, Golang, C, SQL, JavaScript/TypeScript, HTML/CSS

Frameworks/Methodologies: Agile, CI/CD, React, Node.js, Express.js, Next.js, TensorFlow, Scikit-learn

Tools/Platforms: Git, Docker, AWS, Supabase, Firebase, Redis, Claude

Certifications: AWS Certified Cloud Practitioner (2025), SAS Base Programming (2022)